Smart Healthcare Network Design

Introduction

You are assigned to design a Smart Healthcare Network connecting multiple hospitals and medical units under one system. The network must support real-time patient data transfer, email communication between hospitals, automated IP assignment, and fault-tolerant routing.

The network includes Central Hospital (CH), Diagnostic Center (DC), Emergency Unit (EU), Research Lab (RL), Pharmacy Hub (PH), and Remote Clinic (RC).

Device Distribution

Central Hospital (CH) - 300 devices
Diagnostic Center (DC) - 150 devices
Emergency Unit (EU) - 80 devices
Research Lab (RL) - 120 devices
Pharmacy Hub (PH) - 100 devices
Remote Clinic (RC) - 60 devices

Requirements

Topology Design & Server Setup

- Each unit must have one router.
- CH acts as the central backbone.
- DC connects only to CH.
- RL, PH, EU connected via a core switch.
- RC connects to both PH and EU for redundancy.
- CH connects directly to RL and EU.
- Each unit has its own Email Server.
- Configure email communication between CH and RL.
- CH hosts DNS and Web server (www.health.local).

DHCP Configuration

- DC and EU routers provide DHCP.

- RL and PH use dedicated DHCP servers.
- CH and RC use static IP configuration.

Addressing

Base network must be generated using the student ID. Take the last 4 digits and split into two parts to form the first two octets. The remaining octets will be 0 with /16 prefix.
Example: If ID is 20234567 → 45.67 → Network: 45.67.0.0/16

Use VLSM to subnet based on requirements in the following order:

- CH - /23 (first subnet)
- DC - /24
- RL - /25
- PH - /25
- EU - /25
- RC - /26 (last subnet)

Routing Design

- CH to DC: Uses next-hop static routing.
- EU to CH: Uses exit-interface static routing.
- PH to DC network: Uses recursive static routing via RL (specific route to reach DC through RL's next-hop).
- RC to CH network: Uses floating static route via EU as backup (with higher administrative distance than RIP-learned routes).
- RL, PH, EU: Use RIPv2 for dynamic routing among themselves and general network connectivity.
- CH: Uses both static routing (to DC) and dynamic routing (RIPv2). Must redistribute static routes into RIPv2 domain.
- Routing must ensure redundancy and avoid loops.

Special Conditions

- If CH-RL link fails, traffic reroutes via EU using RIPv2 convergence.
- RC must always maintain two paths: primary via PH (RIP), backup via EU (floating static route).
- If the primary path from PH to CH (via RL) is unavailable, PH should use the recursive static route through RL as fallback.

Configuration Notes

- Next-hop static routing: Specifies the IP address of the next router.
- Exit-interface static routing: Specifies the outgoing interface instead of next-hop IP.
- Recursive static routing: Points to an intermediate router that knows the path to the final destination.
- Floating static routing: Backup route with higher administrative distance (e.g., 130) that activates when the primary route fails.
- Static and dynamic routing can coexist on the same router for different destination networks.
- Use 'redistribute static' command in RIPv2 configuration on CH to advertise static routes.

Deliverables

- Cisco Packet Tracer (.pkt) file
- Network topology diagram
- VLSM tree
- IP addressing table
- Router configuration commands
- Work distribution